Thai Tax Assistant

Eligibility guidance and a reproducible tax estimate for salary income.

Audience	Scope	Model
Salaried individuals	P.N.D.91 / tax year 2568	gemma-4-E4B-it

Problem

A salary earner needs to determine which payments qualify and how individual and shared ceilings affect the final tax. Eligibility lives in written rules, while computing tax requires consistent arithmetic across several thresholds.

Supported scenario

An individual filer has employment income only and no special income exemptions. The calculation covers the employment expense deduction, personal allowance, eligible own life and health premiums, and eligible Easy E-Receipt 2.0 totals. Withholding is optional and produces a separate balance.

User workflow

The local form accepts annual gross income and eligible amounts. It displays the allowed amounts, the portions that exceed ceilings, taxable net income, and estimated tax before and after the optional items. Chat answers rule questions with links to the relevant manual page and can call the same calculator on a fictional profile.

Example

With 600,000 baht of employment income and eligible life and health premiums of 85,000 and 30,000 baht, the calculator applies 15,000 baht of health premiums after the combined insurance ceiling. The estimate changes from 21,500 to 11,500 baht.

Revenue Department instructions, PDF page 7

System design

A rule retriever supplies evidence and Python computes the numerical result.

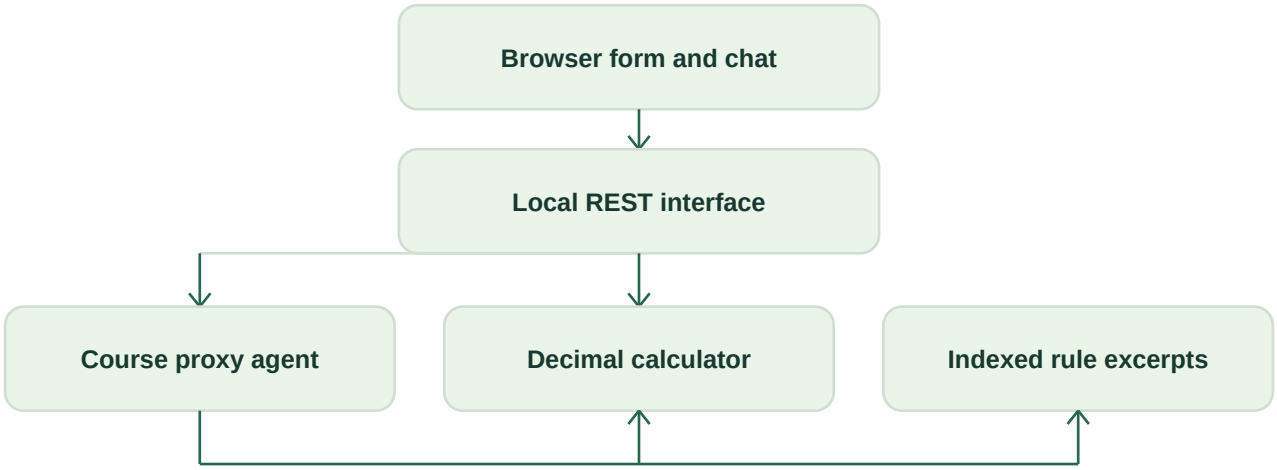

Figure 1. Direct requests call the calculator. Chat uses an agent that can call the calculator and rule retriever.

Grounded retrieval and structured inputs

The corpus contains eleven Thai rule summaries with page links to the 2568 filing instructions. The hybrid retriever combines word overlap and sparse character TF-IDF. Pydantic validates the year, allowed fields and nonnegative baht amounts before calculations.

Agent and application boundaries

The model has two allow-listed tools and a five-round limit. Search results and calculation sources are returned separately from free text. The app treats eligible receipt totals as separate groups and applies the shared insurance ceiling in code. The browser warns that chat is sent to the class proxy. The direct form calculates locally.

Evaluation

Thirty labelled cases separate evidence retrieval from arithmetic and model extraction.

Offline benchmark

Fifteen Thai rule questions identify their expected source entries. Recall@3 is the average share of expected entries in the first three hits. The keyword baseline uses word overlap over title and text. The hybrid adds sparse character vectors and curated synonyms. The comparison measures those changes together.

Offline retrieval	Cases	Recall@3
Keyword overlap	15	0.3333
Hybrid	15	0.9333

The direct calculator matched 15 of 15 independently calculated profiles. Cases include exact bracket boundaries, amounts above a deduction ceiling, the special receipt tier, and a withholding balance.

Class proxy comparison

Live method	Exact tax before withholding
Model only	1 / 15
Model with tools	15 / 15

Both methods receive the same synthetic profile and are scored on the tax before withholding. The baseline finished and yielded a readable number on every case. An absent or malformed result would count as a miss.

Reproduce from the repository root with python -m tax_app.evaluate. Add --live to score the configured course proxy.

Errors and limitations

The benchmark exposes which failures remain possible even with correct arithmetic.

Observed retrieval failure

The hybrid missed the exclusion rule for a monthly internet bill. The expected rule ID was **receipt-excluded**, while the top results were expense, personal, receipt-period. The exact prompt and ranked IDs are recorded in the checked-in results.

Supported assumptions

This project estimates one tax year and selected deductions. It assumes the user has verified the eligibility of amounts entered into the calculator. The tax figure can change with additional deductions or special exemptions. The chat model can misread units or omit information even when the underlying arithmetic is correct.

Next work

Collect held-out Thai paraphrases and label the actual eligibility answer separately from retrieval. Add confirmation of the extracted profile before calculation. A future rule version could support another tax year with a separate corpus and test set.

Official sources

Revenue Department filing materials for 2568

P.N.D.91 instructions, including deductions and bracket table

P.N.D.91 calculation form, including withholding and balance